

Project Report: 5G Network Performance Analysis

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1. Executive Summary

According to this analysis, 5G network performance is very constant and calls for a comprehensive, multi-indicator monitoring strategy as opposed to depending just on single-factor measures.

2. Data Preparation

- **Methodology:** Orchestrated a robust Python environment to process 50,000 network logs.
- **Validation:** Confirmed 0% error rate for missing values and duplicates, ensuring high-quality data integrity.

3. Exploratory Data Analysis (EDA)

- **Data Profile:** Histogram analysis demonstrates a balanced data usage distribution (0–500 MB).
- **Performance Parity:** The average latency (10–11 ms) and download throughput (~550 Mbps) are the same for 4G, 5G NSA, and 5G SA architectures.
- **Variable Independence:** Heatmap analysis confirms weak correlation between isolated metrics like signal strength and overall network performance.

4. Analysis & Actionable Insights

- **Congestion Findings:** Throughput remains stable regardless of traffic density, challenging traditional congestion assumptions.

- **Strategic Recommendations:** To increase overall network reliability, telco providers should switch to integrated dashboards that incorporate speed, latency, traffic density, and signal quality.

5. Project Demonstration

- **Presentation Video:** <https://youtu.be/LQWTVRfmLHY?si=SMoi-JMGQe5Op66I>
- **GitHub Repository:** <https://github.com/AkifDzikri/5G-Network-Performance-Analysis>